

```
In [76]: import sqlite3
import nfl_data_py as nfl
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Connect to your "filing cabinet"
conn = sqlite3.connect('nfl_data.db')
```

```
In [77]: query = """
SELECT
    play_type,
    COUNT(*) as total_plays,
    AVG(epa) as avg_epa,
    AVG(yards_gained) as avg_yards
FROM pbp_data
WHERE posteam = 'IND'
    AND season = 2025
    AND down IN (1,2)
    AND play_type IN ('run','pass')
GROUP BY play_type
"""

df_colts = pd.read_sql_query(query, conn)
print("Colts Early Down Efficiency (2025):")
print(df_colts)
```

```
Colts Early Down Efficiency (2025):
  play_type  total_plays  avg_epa  avg_yards
0      pass           423  0.043757   6.520095
1       run           361  0.051760   4.775623
```

```
In [78]: query = """
SELECT
    CASE
        WHEN week <= 8 THEN 'Weeks 1-8'
        WHEN week BETWEEN 9 AND 13 THEN 'Weeks 9-13'
        ELSE 'Other'
    END AS season_segment,
    SUM(CASE WHEN play_type = 'run' THEN 1 ELSE 0 END) AS run_plays,
    SUM(CASE WHEN play_type = 'pass' THEN 1 ELSE 0 END) AS pass_plays,
    CAST(SUM(CASE WHEN play_type = 'run' THEN 1 ELSE 0 END) AS FLOAT) /
    NULLIF(SUM(CASE WHEN play_type = 'pass' THEN 1 ELSE 0 END), 0) AS run_to_pa
FROM pbp_data
WHERE posteam = 'IND'
    AND season = 2025
    AND down IN (1)
    AND play_type IN ('run', 'pass')
    AND week <= 13
GROUP BY season_segment
ORDER BY season_segment;
"""
```

```
df_season_segments = pd.read_sql_query(query, conn)
print(df_season_segments)
```

	season_segment	run_plays	pass_plays	run_to_pass_ratio
0	Weeks 1-8	107	115	0.930435
1	Weeks 9-13	47	59	0.796610

```
In [82]: query_1st = """
SELECT
    CASE WHEN week <= 8 THEN 'Early (1-8)' ELSE 'Late (9-13)' END as season_segment
    CAST(SUM(CASE WHEN play_type = 'run' THEN 1 ELSE 0 END) AS FLOAT) /
    NULLIF(SUM(CASE WHEN play_type = 'pass' THEN 1 ELSE 0 END), 0) as run_to_pass_r
FROM pbp_data
WHERE posteam = 'IND' AND season = 2025 AND down = 1 AND play_type IN ('run', 'pass')
GROUP BY season_segment
"""

df_1st = pd.read_sql_query(query_1st, conn)
print("--- 1st Down Run/Pass Ratios ---")
print(df_1st)
```

```
--- 1st Down Run/Pass Ratios ---
   season_segment  run_to_pass_ratio
0      Early (1-8)             0.930435
1      Late (9-13)             1.081081
```

```
In [81]: query = """
SELECT
    CASE WHEN week <= 8 THEN 'Early (1-8)' ELSE 'Late (9-13)' END as season_segment
    play_type,
    COUNT(*) as total_plays,
    AVG(epa) as avg_epa,
    AVG(yards_gained) as avg_yards
FROM pbp_data
WHERE posteam = 'IND'
    AND season = 2025
    AND down IN (1)
    AND play_type IN ('run', 'pass')
GROUP BY season_segment, play_type
ORDER BY season_segment, play_type
"""

df_colts_split = pd.read_sql_query(query, conn)
print("Colts 1st Down Run Pass Ratio: Early Weeks vs. Rest of Season")
print(df_colts_split)
```

```
Colts 1st Down Run Pass Ratio: Early Weeks vs. Rest of Season
   season_segment play_type  total_plays  avg_epa  avg_yards
0      Early (1-8)      pass         115  0.292449   8.660870
1      Early (1-8)      run          107  0.119862   6.093458
2      Late (9-13)      pass         111 -0.185216   5.882883
3      Late (9-13)      run          120 -0.108230   3.541667
```

```
In [84]: import nfl_data_py as nfl
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```

# 1. Load Data
pbp_data = nfl.import_pbp_data([2025])
pbp_data['season_segment'] = pbp_data['week'].apply(lambda x: 'Early (1-8)' if x <=

# 2. Process Data
colts_comparison = pbp_data[
    (pbp_data['posteam'] == 'IND') &
    (pbp_data['season'] == 2025) &
    (pbp_data['down'].isin([1])) &
    (pbp_data['play_type'].isin(['run', 'pass']))
].groupby(['season_segment', 'play_type']).agg({
    'play_id': 'count',
    'epa': 'mean'
}).rename(columns={'play_id': 'total_plays', 'epa': 'avg_epa'}).reset_index()

# 3. Visualize
plt.figure(figsize=(10, 6))
sns.barplot(
    data=colts_comparison,
    x="season_segment",
    y="avg_epa",
    hue="play_type"
)

plt.title("Colts Efficiency Collapse: EPA by Season Segment")
plt.ylabel("Average EPA")
plt.xlabel("Season Period")
plt.show()

```

2025 done.

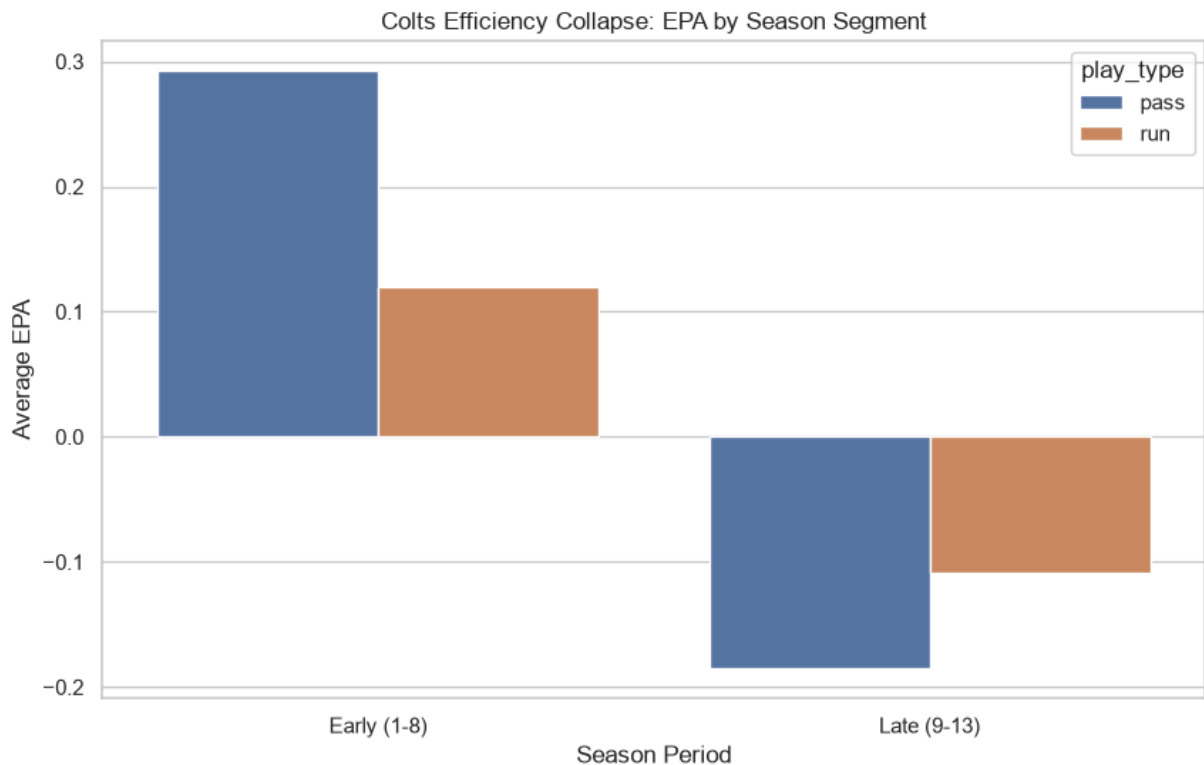
Downcasting floats.

C:\Users\camer\AppData\Local\Temp\ipykernel_18396\4163263661.py:8: PerformanceWarning: DataFrame is highly fragmented. This is usually the result of calling `frame.insert` many times, which has poor performance. Consider joining all columns at once using `pd.concat(axis=1)` instead. To get a de-fragmented frame, use `newframe = frame.copy()`

```

pbp_data['season_segment'] = pbp_data['week'].apply(lambda x: 'Early (1-8)' if x <
= 8 else 'Late (9-13)')

```



```
In [80]: query_2nd = """
SELECT
    CASE WHEN week <= 8 THEN 'Early (1-8)' ELSE 'Late (9-13)' END as season_segment
    CAST(SUM(CASE WHEN play_type = 'run' THEN 1 ELSE 0 END) AS FLOAT) /
    NULLIF(SUM(CASE WHEN play_type = 'pass' THEN 1 ELSE 0 END), 0) as run_to_pass_r
FROM pbp_data
WHERE posteam = 'IND' AND season = 2025 AND down = 2 AND play_type IN ('run', 'pass')
GROUP BY season_segment
"""

df_2nd = pd.read_sql_query(query_2nd, conn)
print("--- 2nd Down Run/Pass Ratios ---")
print(df_2nd)
```

```
--- 2nd Down Run/Pass Ratios ---
  season_segment  run_to_pass_ratio
0    Early (1-8)             0.765432
1    Late (9-13)             0.620690
```

```
In [79]: query = """
SELECT
    CASE WHEN week <= 8 THEN 'Early (1-8)' ELSE 'Late (9-13)' END as season_segment
    play_type,
    COUNT(*) as total_plays,
    AVG(epa) as avg_epa,
    AVG(yards_gained) as avg_yards
FROM pbp_data
WHERE posteam = 'IND'
    AND season = 2025
    AND down IN (2)
    AND play_type IN ('run', 'pass')
GROUP BY season_segment, play_type
ORDER BY season_segment, play_type
```

```
"""
```

```
df_colts_split = pd.read_sql_query(query, conn)
print("Colts 2nd Down Run Pass Ratio: Early Weeks vs. Rest of Season")
print(df_colts_split)
```

Colts 2nd Down Run Pass Ratio: Early Weeks vs. Rest of Season

	season_segment	play_type	total_plays	avg_epa	avg_yards
0	Early (1-8)	pass	81	0.186299	5.790123
1	Early (1-8)	run	62	0.211304	4.596774
2	Late (9-13)	pass	116	-0.083222	5.517241
3	Late (9-13)	run	72	0.079817	5.027778

```
In [83]: # 1. Load Data
pbp_data = nfl.import_pbp_data([2025])
pbp_data['season_segment'] = pbp_data['week'].apply(lambda x: 'Early (1-8)' if x <=

# 2. Process Data
colts_comparison = pbp_data[
    (pbp_data['posteam'] == 'IND') &
    (pbp_data['season'] == 2025) &
    (pbp_data['down'].isin([2])) &
    (pbp_data['play_type'].isin(['run', 'pass']))
].groupby(['season_segment', 'play_type']).agg({
    'play_id': 'count',
    'epa': 'mean'
}).rename(columns={'play_id': 'total_plays', 'epa': 'avg_epa'}).reset_index()

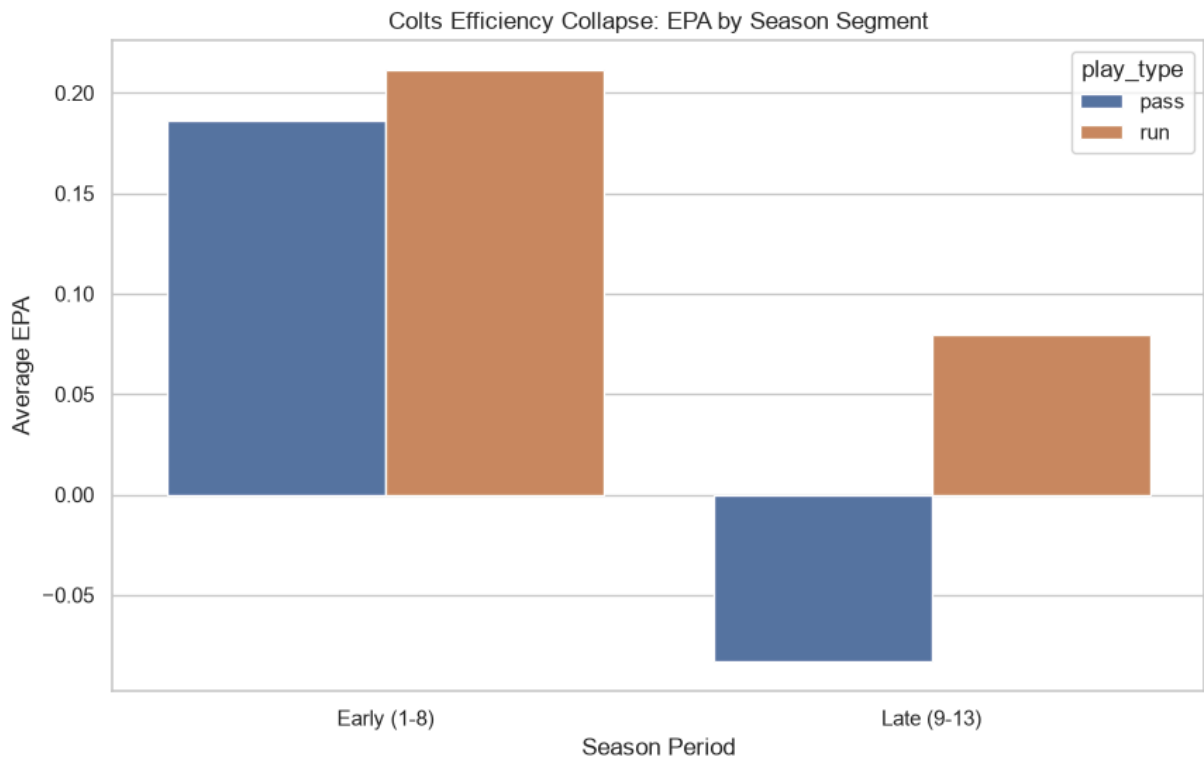
# 3. Visualize
plt.figure(figsize=(10, 6))
sns.barplot(
    data=colts_comparison,
    x="season_segment",
    y="avg_epa",
    hue="play_type"
)

plt.title("Colts Efficiency Collapse: EPA by Season Segment")
plt.ylabel("Average EPA")
plt.xlabel("Season Period")
plt.show()
```

2025 done.

Downcasting floats.

```
C:\Users\camer\AppData\Local\Temp\ipykernel_18396\4161827230.py:3: PerformanceWarning:
DataFrame is highly fragmented. This is usually the result of calling `frame.insert`
many times, which has poor performance. Consider joining all columns at once u
sing pd.concat(axis=1) instead. To get a de-fragmented frame, use `newframe = frame.
copy()`
    pbp_data['season_segment'] = pbp_data['week'].apply(lambda x: 'Early (1-8)' if x <
= 8 else 'Late (9-13)')
```



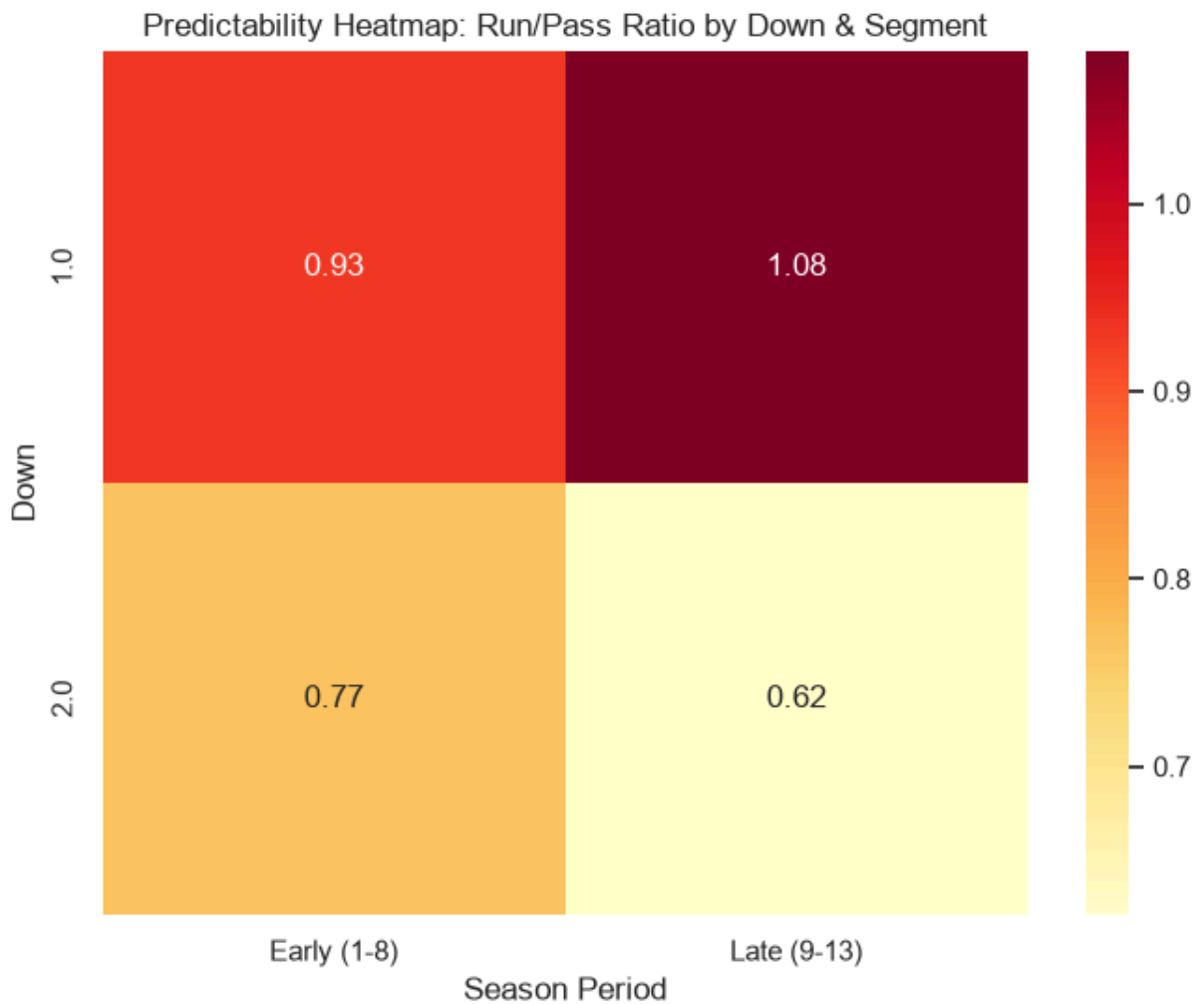
```
In [85]: # 1. Filter and Aggregate: Calculate counts of run/pass for each down and segment
heatmap_data = pbp_data[
    (pbp_data['posteam'] == 'IND') &
    (pbp_data['season'] == 2025) &
    (pbp_data['down'].isin([1, 2])) &
    (pbp_data['play_type'].isin(['run', 'pass']))
].groupby(['season_segment', 'down', 'play_type']).size().unstack(fill_value=0)

# 2. Calculate the Ratio
# (This acts Like a SQL calculation: CASE WHEN...)
heatmap_data['ratio'] = heatmap_data['run'] / heatmap_data['pass']

# 3. Pivot for Heatmap format
# Rows: Down | Columns: Season Segment
heatmap_pivot = heatmap_data.reset_index().pivot(index='down', columns='season_segmen

# 4. Visualize
plt.figure(figsize=(8, 6))
sns.heatmap(
    heatmap_pivot,
    annot=True,
    cmap="YlOrRd", # Yellow to Red (Red = higher predictability/higher run ratio)
    fmt=".2f"
)

plt.title("Predictability Heatmap: Run/Pass Ratio by Down & Segment")
plt.ylabel("Down")
plt.xlabel("Season Period")
plt.show()
```



```
In [86]: # Query for "Manageable" 2nd Downs (1-6 yards to go)
query_manageable = """
SELECT
    CASE WHEN week <= 8 THEN 'Early (1-8)' ELSE 'Late (9-13)' END as season_segment
    play_type,
    COUNT(*) as total_plays,
    AVG(epa) as avg_epa
FROM pbp_data
WHERE posteam = 'IND'
    AND season = 2025
    AND down = 2
    AND ydstogo <= 6
    AND play_type IN ('run', 'pass')
GROUP BY season_segment, play_type
"""

df_manageable = pd.read_sql_query(query_manageable, conn)
print(df_manageable)
```

	season_segment	play_type	total_plays	avg_epa
0	Early (1-8)	pass	16	0.106199
1	Early (1-8)	run	45	0.312252
2	Late (9-13)	pass	30	-0.179705
3	Late (9-13)	run	50	0.138975

```
In [88]: # 1. Define Success Logic based on your "4-yard" rule
# We add this column directly to your pbp_data
pbp_data['is_success'] = (
    ((pbp_data['down'] == 1) & (pbp_data['yards_gained'] >= 4)) |
    ((pbp_data['down'] == 2) & (pbp_data['yards_gained'] >= (pbp_data['ydstogo'] /
    ).astype(int))

# 2. Groupby for the "Forensic" table
# We group by segment and play_type, then aggregate success rate and EPA
forensic_table = pbp_data[
    (pbp_data['posteam'] == 'IND') &
    (pbp_data['season'] == 2025) &
    (pbp_data['down'].isin([1, 2])) &
    (pbp_data['play_type'].isin(['run', 'pass']))
].groupby(['season_segment', 'play_type']).agg(
    total_plays=('play_type', 'size'),
    success_rate_pct=('is_success', lambda x: x.mean() * 100),
    avg_epa=('epa', 'mean')
).reset_index()

print(forensic_table)
```

	season_segment	play_type	total_plays	success_rate_pct	avg_epa
0	Early (1-8)	pass	196	60.204082	0.248581
1	Early (1-8)	run	169	55.029586	0.153409
2	Late (9-13)	pass	227	46.696035	-0.133096
3	Late (9-13)	run	192	50.520833	-0.037713

C:\Users\camer\AppData\Local\Temp\ipykernel_18396\2227653607.py:3: PerformanceWarning: DataFrame is highly fragmented. This is usually the result of calling `frame.insert` many times, which has poor performance. Consider joining all columns at once using `pd.concat(axis=1)` instead. To get a de-fragmented frame, use `newframe = frame.copy()`

```
pbp_data['is_success'] = (
```

```
In [89]: # 1. Define segments and filter for the 1-13 regular season window
# This ensures no "future" data (post-week 13 or post-season) leaks into your calcu
pbp_1_13 = pbp_data[
    (pbp_data['season'] == 2025) &
    (pbp_data['season_type'] == 'REG') &
    (pbp_data['week'] <= 13)
].copy()

# Create the segment label
pbp_1_13['segment'] = pd.cut(
    pbp_1_13['week'],
    bins=[0, 8, 13],
    labels=['Early (1-8)', 'Late (9-13)']
)

# 2. Calculate Defensive Strength for EACH segment independently
# We define "strength" as the average EPA allowed by that defense during that speci
def_strength_early = pbp_1_13[pbp_1_13['segment'] == 'Early (1-8)'].groupby('deftea
def_strength_late = pbp_1_13[pbp_1_13['segment'] == 'Late (9-13)'].groupby('deftea

# Combine strengths into a mapping table
```



```

strength_map = pd.concat([def_strength_early, def_strength_late])

# 3. Analyze performance and merge with difficulty
# Grouping by week and team ensures the result is chronologically ordered
df_final = pbp_1_13[pbp_1_13['posteam'] == 'IND'].copy()
df_final = df_final.merge(strength_map, on='defteam', how='left')

comparison_table = df_final.groupby(['segment', 'week', 'defteam']).agg(
    colts_vs_them=('epa', 'mean'),
    opp_strength=('opp_def_strength', 'mean')
).sort_index(level=['segment', 'week'])

print(comparison_table)

```

segment	week	defteam	colts_vs_them	opp_strength
Early (1-8)	1	MIA	0.251387	0.045084
	2	DEN	0.155293	-0.052225
	3	TEN	0.266243	0.048793
	4	LA	-0.063891	-0.074305
	5	LV	0.242410	0.009166
	6	ARI	0.166566	0.027992
	7	LAC	0.372199	-0.043899
	8	TEN	0.296975	0.048793
Late (9-13)	9	PIT	-0.041573	0.018146
	10	ATL	0.104444	0.020964
	12	KC	0.013101	0.058511
	13	HOU	-0.052534	-0.107708

```

In [90]: # 1. Setup and Style
# Bar charts benefit from a horizontal grid for easier level comparison.
plt.figure(figsize=(14, 7))
sns.set_style("whitegrid")
plt.title('2025 Colts: Situational Performance vs. Opponent Difficulty Audit', font

# 2. Data Preparation
# Crucial step: Reset index so 'week' and 'segment' are columns, not index levels.
viz_df = comparison_table.reset_index()

# Sort by week to ensure chronological order.
viz_df = viz_df.sort_values('week')

# Melt the dataframe into long format. This allows Seaborn to easily plot grouped b
# It transforms the two EPA columns into one 'EPA' column and one 'Metric Type' Lab
viz_melted = viz_df.melt(
    id_vars=['week', 'segment', 'defteam'],
    value_vars=['colts_vs_them', 'opp_strength'],
    var_name='Metric Type',
    value_name='EPA'
)

# Rename the metrics to be human-readable in the legend.
viz_melted['Metric Type'] = viz_melted['Metric Type'].map({
    'colts_vs_them': 'Colts Performance (vs. Them)',
    'opp_strength': 'Opponent Difficulty (Defensive Avg.)'
})

```

```

# 3. Plotting the Bars
# Grouped bar chart with clean, distinct colors.
# Note: 'dodge=True' is the default but crucial for grouping.
sns.barplot(
    data=viz_melted,
    x='week',
    y='EPA',
    hue='Metric Type',
    palette=['#1a437c', '#e0e0e0'], # Direct comparison: bold blue vs. muted gray
    edgecolor='black',
    linewidth=1
)

# 4. Context & Analysis Annotation
# Add the vertical dashed line at Week 8.
plt.axvline(x=7.5, color='#d9534f', linestyle='--', linewidth=1.5)

# Add shading and labels to define the two segments.
plt.axvspan(7.5, 12.5, color='#d9534f', alpha=0.10) # Shade Late segment light red
plt.axvspan(-0.5, 7.5, color='#e6e6e6', alpha=0.15) # Optional grey background for

# Centered Labels within each region.
plt.text(3.5, plt.gca().get_ylim()[1] * 0.9, 'Early (1-8)', fontsize=12, horizontal
plt.text(10, plt.gca().get_ylim()[1] * 0.9, 'Late (9-13): Regression Identified', f

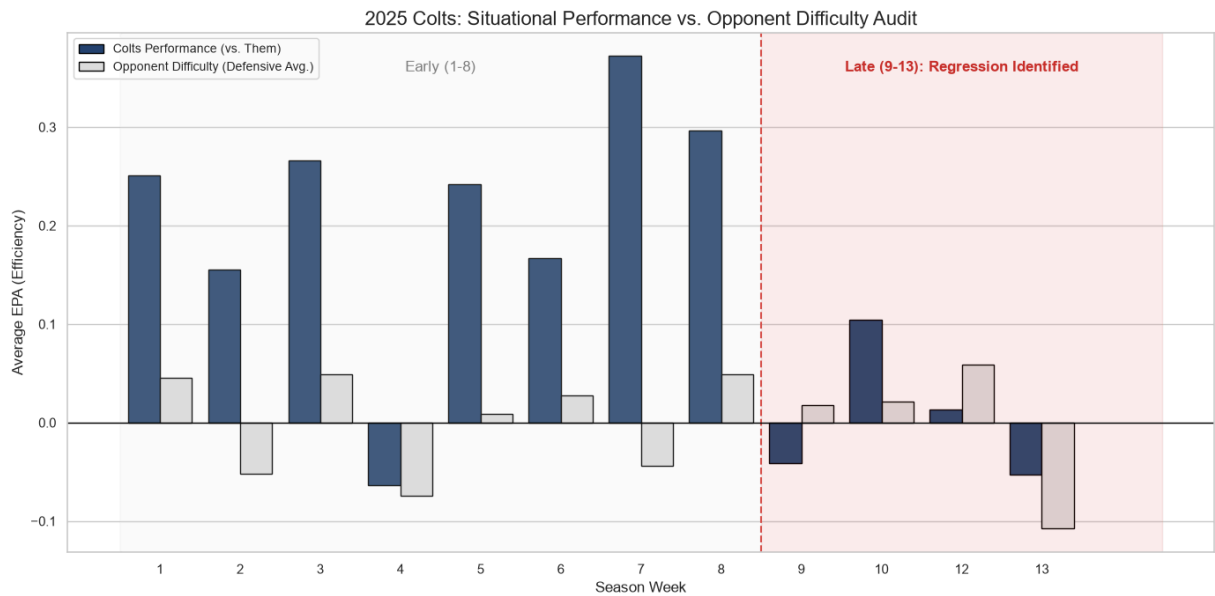
# 5. Fine-Tuning the Axes
plt.xlabel('Season Week', fontsize=12)
plt.ylabel('Average EPA (Efficiency)', fontsize=12)

# Centering the zero Line (which represents League average)
plt.axhline(0, color='black', linewidth=1)

# Ensure the Legend is informative but unobtrusive.
plt.legend(frameon=True, fontsize=10, loc='best')
plt.tight_layout()

plt.show()

```



```
In [87]: # Calculate Success Rate: A positive EPA play is "successful"
df_success = pbp_data[
    (pbp_data['posteam'] == 'IND') &
    (pbp_data['down'] == 1)
].copy()

df_success['is_success'] = (df_success['epa'] > 0).astype(int)

# Group by Segment to see if the "Success" of their strategy actually changed
success_analysis = df_success.groupby('season_segment')['is_success'].mean()
print(success_analysis)
```

```
season_segment
Early (1-8)    0.487395
Late (9-13)    0.378049
Name: is_success, dtype: float64
```